Rami Abou-Shamalah (P.Geo)

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Education

MS, Applied Data Science 2023-2025
University of Michigan

BSc, Applied Mathematics & Geology 2016-2020
University of Western Ontario

Experience

Data Geoscientist June 2023 - Present
Sander Geophysics *Ottawa, Canada*

- Built end-to-end parallelized **data pipeline** that extracts historical projects from on-premise servers, fetching data within custom databases, performs QA/QC, removes outliers, normalizes the data, and engineers features - automating decades worth of training data for **machine learning training** purposes.
- Implemented state of the art time series anomaly detection model (AnomalyTransformer) via Arxiv in **PyTorch**, incorporated **WandB** to track experiments in real-time and Hydra for configuration file management. This model ultimately **reduced processing time by over 65%** by identifying anomalies.
- Developed **Python programs** leveraging NumPy and Pandas for conducting statistical analysis of flight data by assessing the quality against client specifications, resulting in **$10,000 in annual savings**.
- Processed airborne geophysical data including: Magnetic, Radiometric, and Gravity.
- Specialized in the delineation and interpretation of geological phenomena and prospective mineralization targets within the combination of geophysical, geological, geochemical, and satellite datasets.

Data Analyst / Engineer June 2022 - Present
1Stop *Remote*

- Provided executives with a centralized dashboard for real-time, data-driven decision making, architecting and deploying a **data warehouse in SnowFlake**, consolidating disparate sources including: payroll, ad spend, web analytics, and revenue streams.
- Optimized caching and automated **ETL flows from Firebase into IndexedDB** (via REST APIs), allowing caching of 500MB+ of data client-side, reducing **API calls by 85%.**
- Built OKR dashboards to **quantify marketing ROI** volume for stakeholder meetings, directly proving ad spend increases appointments volume.
- Developed a revenue **forecasting model in Python** using historical data, achieving 85% accuracy.
- Embedded Google Maps API to enable geospatial analysis, guiding city-level marketing decisions.
- Maintained **internal API documentation** and built reusable RESTful endpoints for data ingestion.

Data Analyst December 2021 – June 2023
Innovative Mining Solutions *London, Canada*

- Led team of 5 in **data collection and curation** process: overseeing collection, digitizing, cleaning, transforming, and implementing QA/QC, thus preparing high quality and **ML-ready training data**.
- Engineered geospatial features using QGIS software **boosting model precision by ~22%.**
- Presented domain-specifics for non-technical executives in company-wide meetings.

Projects

Geospatial Prospectivity AI Platform 2025

- Designed a **geospatial AI platform** for mineral prospectivity using CNN-based analysis on geophysical raster datasets (gravity, magnetic). Trained **deep models** to identify mineral potential hotspots.
- Served user-selected regions via **Supabase API, using spatial queries** to dynamically return results. Interactive Mode